

Equal Areas of Vertex Triangles in a Regular Polygon

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Abstract

Let $a(n)$ be the number of distinct areas of triangles whose vertices are vertices of a regular n -gon. We obtain a closed formula for $a(n)$. The proof reduces equality of areas to a conjugation-stable vanishing relation of twelve roots of unity and then applies the classification of Poonen and Rubinstein together with a short computer-assisted exact case analysis.

1 Statement

Choose three vertices of a regular n -gon. Let r, s, t be the numbers of polygon sides in the three arcs between consecutive chosen vertices. Then

$$r, s, t > 0, \quad r + s + t = n.$$

The unordered triple (r, s, t) determines the triangle up to congruence. Hence the number of triangle shapes is the number of partitions of n into three positive parts,

$$p_3(n) = \text{round}\left(\frac{n^2}{12}\right) = \text{A069905}(n).$$

We now determine how many of these triangle shapes have distinct areas.

Theorem. *For $n \geq 3$,*

$$a(n) = \text{round}\left(\frac{n^2}{12}\right) - \mathbf{1}_{4|n} \left\lfloor \frac{n-4}{8} \right\rfloor - \mathbf{1}_{24|n} - 2\mathbf{1}_{30|n}.$$

Thus two noncongruent vertex triangles of a regular n -gon have equal area if and only if $n \geq 12$ and $4 \mid n$, or $30 \mid n$.

2 Reduction to roots of unity

If the circumradius is R , the triangle represented by (r, s, t) has area

$$2R^2 \sin \frac{\pi r}{n} \sin \frac{\pi s}{n} \sin \frac{\pi t}{n}.$$

Put

$$u = \frac{r}{n}, \quad v = \frac{s}{n}, \quad w = \frac{t}{n}.$$

Two shapes have equal area exactly when

$$\sin(\pi u) \sin(\pi v) \sin(\pi w) = \sin(\pi x) \sin(\pi y) \sin(\pi z), \quad (1)$$

where all six numbers are positive rationals and

$$u + v + w = x + y + z = 1. \quad (2)$$

Since $A + B + C = \pi$ implies

$$4 \sin A \sin B \sin C = \sin(2A) + \sin(2B) + \sin(2C),$$

(1) is equivalent to

$$\sin(2\pi u) + \sin(2\pi v) + \sin(2\pi w) = \sin(2\pi x) + \sin(2\pi y) + \sin(2\pi z). \quad (3)$$

Define

$$\begin{aligned} \alpha_1 &= \frac{1}{2} - 2u, & \alpha_2 &= \frac{1}{2} - 2v, & \alpha_3 &= \frac{1}{2} - 2w, \\ \alpha_4 &= \frac{3}{2} - 2x, & \alpha_5 &= \frac{3}{2} - 2y, & \alpha_6 &= \frac{3}{2} - 2z. \end{aligned}$$

Then (3) becomes

$$\sum_{j=1}^6 e^{i\pi\alpha_j} + \sum_{j=1}^6 e^{-i\pi\alpha_j} = 0, \quad (4)$$

and (2) gives

$$\sum_{j=1}^6 \alpha_j = 2. \quad (5)$$

Thus an equal-area pair gives a conjugation-stable vanishing relation of weight 12 among roots of unity.

Poonen and Rubinstein prove that their Table 1 is a complete list, up to rotation, of the minimal vanishing relations of weight at most 12 [1, Theorem 3]. Their Table 2 lists all decomposition types of a relation of weight 12, and their Lemma 4 shows that a conjugation-stable relation of weight at most 12 has a conjugation-stable decomposition into minimal relations [1].

3 The finite cases

The companion program `verify_classification.py` reconstructs the relations in Table 1 from the notation of Poonen and Rubinstein and reproduces, in weight 12, the numbers

$$3, 15, 36, 14, 1.$$

It then applies (4), (5), positivity, and the two sum conditions in (2) to the first 22 decomposition types in Table 2, using exact rational arithmetic. Apart from permutations and interchange of the two triples, exactly three nontrivial solutions remain:

$$\left(\frac{1}{12}, \frac{5}{12}, \frac{1}{2}\right) \sim \left(\frac{1}{8}, \frac{1}{4}, \frac{5}{8}\right), \quad (6)$$

$$\left(\frac{1}{30}, \frac{11}{30}, \frac{3}{5}\right) \sim \left(\frac{1}{15}, \frac{1}{5}, \frac{11}{15}\right), \quad (7)$$

and

$$\left(\frac{2}{15}, \frac{2}{5}, \frac{7}{15}\right) \sim \left(\frac{1}{5}, \frac{7}{30}, \frac{17}{30}\right). \quad (8)$$

Here $\sim$ denotes equality of areas. The program uses no floating-point arithmetic.

The remaining decomposition type in Table 2 is $6R_2$. It can be handled directly. In an R_2 decomposition each root is paired with its negative. For two variables on the same side of (1), such a pairing gives either a difference of $1/2$ or a sum of 1; the latter is impossible because the third member of that triple is positive. For variables on opposite sides it gives

$$u = x, \quad u + x = \frac{1}{2}, \quad \text{or} \quad u + x = \frac{3}{2}. \quad (9)$$

A root pair can also pair with itself only when the corresponding variable is $1/2$.

If a self-pair occurs, one variable on each side is $1/2$; the remaining two variables on each side sum to $1/2$, and (9) gives the same unordered triple on both sides. If there are no same-side pairs, all pairs are mixed. Summing the relations in (9) shows again that the two unordered triples are equal: a $u + x = 1/2$ pair either fixes $u = x = 1/4$ or occurs with its complementary partner, while a $u + x = 3/2$ pair can only give the same value on the two sides.

Hence a nontrivial solution must have one same-side pair on each side and one mixed pair. Write the two same-side pairs as

$$t, t + \frac{1}{2} \quad \text{and} \quad s, s + \frac{1}{2}.$$

The remaining entries are $1/2 - 2t$ and $1/2 - 2s$. If these are equal, the two triples are the same. Their sum cannot be $3/2$, so the only nontrivial possibility from (9) is

$$\left(\frac{1}{2} - 2t\right) + \left(\frac{1}{2} - 2s\right) = \frac{1}{2},$$

that is, $s = 1/4 - t$. We obtain the single family

$$\boxed{\left(t, \frac{1}{2} - 2t, \frac{1}{2} + t\right) \sim \left(2t, \frac{1}{4} - t, \frac{3}{4} - t\right), \quad 0 < t < \frac{1}{4}.} \quad (10)$$

At $t = 1/8$ the two unordered triples coincide.

4 Counting the collisions

For a regular n -gon, put $t = k/n$ in (10). The terms containing $1/4$ force $4 \mid n$. Write $n = 4m$. Then (10) becomes

$$(k, 2m - 2k, 2m + k) \sim (2k, m - k, 3m - k). \quad (11)$$

The values k and $m - k$ give the same pair, and $k = m/2$ is the trivial fixed point when it is integral. Thus (11) gives

$$\left\lfloor \frac{m-1}{2} \right\rfloor = \left\lfloor \frac{n-4}{8} \right\rfloor$$

nontrivial equal-area pairs.

Solution (6) occurs exactly when $24 \mid n$ and gives, for $n = 24h$,

$$(2h, 10h, 12h) \sim (3h, 6h, 15h). \quad (12)$$

Solutions (7) and (8) occur exactly when $30 \mid n$ and give, for $n = 30h$,

$$(h, 11h, 18h) \sim (2h, 6h, 22h), \quad (13)$$

$$(4h, 12h, 14h) \sim (6h, 7h, 17h). \quad (14)$$

The pairs in (11)–(14) are disjoint. In (12), the shape $(1/8, 1/4, 5/8)$ is the trivial fixed point $t = 1/8$ of (10); the other shape is not in that family. None of the four normalized shapes in (13)–(14) belongs to (10), and the two pairs are disjoint. Therefore every listed pair reduces the number of distinct areas by exactly one. Subtracting these pairs from $p_3(n)$ proves the theorem.

5 Verification

The verification programs are provided with this note.

- `verify_classification.py` performs the exact finite filter described above. It reconstructs Table 1, confirms that Table 2 has 23 types, and returns exactly the three sporadic pairs (6)–(8) from the first 22 types.
- `verify_data.py` independently represents the sine sums in cyclotomic fields. It verifies the formula exactly for every $3 \leq n \leq 300$ and for selected larger values, and checks through $n = 10000$ that the collision families (11)–(14) are pairwise disjoint and have the claimed total size.

References

- [1] B. Poonen and M. Rubinstein, *The Number of Intersection Points Made by the Diagonals of a Regular Polygon*, SIAM J. Discrete Math. **11** (1998), 135–156. doi:10.1137/S0895480195281246.
- [2] N. J. A. Sloane et al., OEIS A069905, *Number of partitions of n into 3 positive parts*, including the interpretation as the number of incongruent triangles formed from three vertices of a regular n -gon. oeis.org/A069905.